

CareerOS - Production Grade Engineering Blueprint (Version 1)

Vision

Build CareerOS as a production-grade AI-powered career platform using enterprise engineering practices from day one. Every service, deployment, and process should reflect real-world production standards.

High Level Architecture

Internet -> CloudFront -> Load Balancer -> API Gateway -> Service Discovery -> Config Server -> Business Microservices -> Kafka -> Databases/Cache/Search -> AI Platform -> Observability.

Core Microservices

1. Auth
2. User
3. Resume
4. Job
5. News
6. Goal
7. Notification
8. Search
9. Analytics
10. Recommendation
11. Media
12. Audit
13. Interview
14. AI Service
15. LangGraph Agent Service
16. Admin
17. Scheduler

Future: Payment, Subscription, Community, Learning, Recruiter, Company, Chat, Video Interview.

Platform Components

Kafka, Redis, PostgreSQL, MongoDB, Elasticsearch, Vector Database, AWS S3, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Jenkins, ArgoCD, Prometheus, Grafana, Loki, Jaeger/OpenTelemetry.

Engineering Principles

- API-first development
- Domain-Driven Design
- Database-per-service
- Event-driven architecture
- Zero shortcuts
- Security by design
- Observability by default
- Automated testing
- CI/CD from day one
- Infrastructure as Code

Roadmap

Phase 0: Product Vision, PRD, NFRs, ADRs, C4 diagrams, Event Storming, DDD.
Phase 1: Engineering Foundation (Repo, branching, CI/CD, Docker, K8s, monitoring, logging).
Phase 2: Auth + User Services.
Phase 3: Resume.
Phase 4: Jobs.
Phase 5: News CMS.
Phase 6: AI Platform.
Phase 7: Notifications.
Phase 8: Search.
Phase 9: Analytics.
Phase 10: AWS Production Deployment.
Phase 11+: Scaling, performance, chaos engineering, disaster recovery.

Production Standards

Every service includes:

Health checks, liveness/readiness probes, structured logging, correlation IDs, tracing, metrics, JWT, RBAC, validation, global exception handling, Flyway, Docker, OpenAPI, unit/integration tests, Testcontainers, rate limiting, circuit breaker, retries, graceful shutdown, audit logging.

Development Workflow

Requirements -> Architecture -> API Design -> Database Design -> Events -> Implementation -> Testing -> Observability -> Security Review -> Deployment -> Monitoring -> Documentation.